Лабораторна робота №7

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Група ІПЗ-21

Варіант 7

Код консольної програми

```
using Lab7Library.Calculations;
using Lab7Library.TextProcessing;
using Lab7Library.Common;


class Program
{
   static void Main()
   {
       const string divider =
"------------------------------------------------------------
-------------------------";

       double radius = 8;
       double height = 5;

       double volume =
MathTools.CalculateCylinderVolume(radius, height);


       Console.WriteLine(divider);
       Console.WriteLine("Fullname: Vlad Semeniuk");
       Console.WriteLine("Group: IPZ-21");
       Console.WriteLine("Variant: 7");
       Console.WriteLine(divider);

       if (double.IsNaN(volume))
       {
           Console.WriteLine("Radius and height are invalid");
       }
       else
       {
           Console.WriteLine(
               $"Radius: {radius:F2}, height: {height:F2}");
           Console.WriteLine($"Cylinder volume: {volume:F2}");
```

```
        }

        Console.WriteLine(divider);

        Console.WriteLine();
        string text1 = "C# Class Library for the 7th variant";
        string text2 = "Hello 12 my na55me is Vla5d and I am a
flutter developer#1";

        // Full path name works even without using keyword.
        string shotText =
            Lab7Library.TextProcessing.TextTools
                .ReplaceDigitsWithAsterisk(text1);


        // With using we could use shorter variant of calling
TextTools methods.
        string longText =
TextTools.ReplaceDigitsWithAsterisk(text2);



        Console.WriteLine($"Text 1: {shotText}");
        Console.WriteLine($"Text 2: {longText}");


        Random random = new Random();
        double randomNum = random.Next();
        Console.WriteLine($"Random number: {randomNum}");
        // bool isPositive =
ValidationTools.IsPositive(randomNum);
        Console.WriteLine($"Here is the
error:\nProgram.cs(56,27): Error CS0122 : 'ValidationTools' is
inaccessible due to its protection level\n Finished building
project: Lab7App. Succeeded: False. Errors: 1. Warnings: 0");
        Console.WriteLine();
        Console.WriteLine("We can use classes with internal
modifier even if we set up reference and using.");
        Console.WriteLine("We can use it only inside that
project where it was created.");
        Console.WriteLine(divider);

    }
}
```

```
namespace Lab7Library.Calculations;



public static class MathTools
{
   public static double CalculateCylinderVolume(
       double radius,
       double height)
   {
       if (radius <= 0 || height <= 0)
       {
           return double.NaN;
       }

       return Math.PI * radius * radius * height;
   }
}
```

```
namespace Lab7Library.Common;


internal static class ValidationTools
{
   public static bool IsPositive(double value)
   {
       return value > 0;
   }
}
```

```
using System.Text;

namespace Lab7Library.TextProcessing;

public static class
   TextTools
{
   public static string ReplaceDigitsWithAsterisk(string text)
   {
       ArgumentNullException.ThrowIfNull(text);

       var sb = new StringBuilder(text.Length);
```

```
        foreach (char symbol in text)
        {
            sb.Append(char.IsDigit(symbol) ? '*' : symbol);
        }

        return sb.ToString();

    }
}
```

ШІ використовувався тільки для ресьорчу.